

4 **(short)** Derive the expression for the gain of the circuit shown in Fig 4. The operational amplifier is ideal. If $R_1 = 1 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, $C_2 = 20 \text{ pF}$, sketch the Bode magnitude plot of the circuit showing its key characteristics. You are not required to use graph paper. [10]

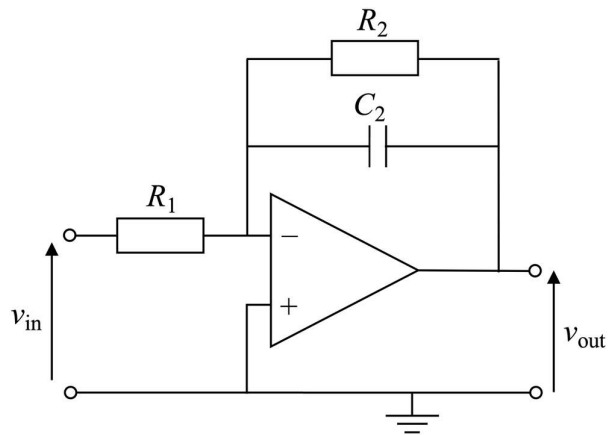


Fig. 4